Record programs:

12. String Palindrome

```
#include <stdio.h>
#include <string.h>

int main() {
   char string1[20];
   int i, length;
   int flag = 0;
   printf("Enter a string: ");
   scanf("%s", string1);
   length = strlen(string1);
   for (i = 0; i < length / 2; i++)
  {
      if (string1[i] != string1[length - i - 1])
          {
          flag = 1;
          break;
           }
   }

   if (flag) {
      printf("%s is not a palindrome\n", string1);
   } else {
      printf("%s is a palindrome\n", string1);
   }

   return 0;
}
```

13. Concatenate Two Strings:

```
#include <stdio.h>
#include <string.h>

int main()
{
   char a[100], b[100];
   printf("Enter the first string\n");
   gets(a);
   printf("Enter the second string\n");
   gets(b)
   strcat(a,b);
```

```
    printf("String obtained on concatenation is %s\n",a);
    return 0;

}
```

14. Simple pointers

```
#include <stdio.h>
int main()
{
  int* ptr, var;

  var = 22;
  printf("Address of var: %p\n", &var);
  printf("Value of var: %d\n\n", var);

  ptr = &c;
  printf("Address of pointer ptr: %p\n", ptr);
  printf("Content of pointer ptr: %d\n\n", *ptr);

  c = 11;
  printf("Address of pointer ptr: %p\n", ptr);
  printf("Content of pointer ptr: %d\n\n", *ptr);

  *ptr = 2;
  printf("Address of var: %p\n", &var);
  printf("Value of var: %d\n\n", var);
  return 0;
}
```

15. Factorial of a number using Void function with arguments

```
#include <stdio.h>

void printFactorial(int n);

int main() {
    int number = 5;
    printFactorial(number);
    return 0;
}

void printFactorial(int n)
{
```

```
        int factorial = 1;
        for (int i = 1; i <= n; i++)
         {
                factorial *= i;
        }

         printf("The factorial of %d is %d\n", n, factorial);
}
```

16. Functions with args and return type

```
#include <stdio.h>
int sumArray(int arr[], int size);

int main()
{
   int numbers[] = {1, 2, 3, 4, 5};
   int size = sizeof(numbers) / sizeof(numbers[0]);
   int total;
   total = sumArray(numbers, size);
   printf("The sum of the array is %d\n", total);
   return 0;
}

int sumArray(int arr[], int size)
{
   int sum = 0;
   for (int i = 0; i < size; i++)
  {
      sum += arr[i];
   }
   return sum;

}
```

17. Fibonacci using Recursion

```
#include <stdio.h>

int fibonacci(int n);

int main()
{
   int n = 10;
```

```
    printf("The %dth Fibonacci number is %d\n", n, fibonacci(n));
    return 0;
}

int fibonacci(int n)
{
    if (n <= 1)
    {
        return n;
    }
    return fibonacci(n - 1) + fibonacci(n - 2);
}
```

18. Swap values using pointers

```
#include <stdio.h>

void swap(int *a, int *b);

int main()
{
    int x = 10;
    int y = 20;
    printf("Before swapping: x = %d, y = %d\n", x, y);
    swap(&x, &y);
    printf("After swapping: x = %d, y = %d\n", x, y);
    return 0;
}

void swap(int *a, int *b)
{
    int temp = *a;
    *a = *b;
    *b = temp;
}
```

19. Structure of Employee

```
#include <stdio.h>
#include <string.h>

struct Employee
{
    int id;
    char name[50];
    float salary;
```

```
};

void displayEmployee(struct Employee emp);

int main()
{
   struct Employee emp1;
   emp1.id = 101;
   strcpy(emp1.name, "John Doe");
   emp1.salary = 75000.50;
   displayEmployee(emp1);
   return 0;
}

void displayEmployee(struct Employee emp) {
   printf("Employee ID: %d\n", emp.id);
   printf("Employee Name: %s\n", emp.name);
   printf("Employee Salary: %.2f\n", emp.salary);
}
```

20. Program to perform file read and write operations

```
#include <stdio.h>

int main() {
   FILE *file;
   char filename[] = "file1.txt";
   file = fopen(filename, "w");
   if (file == NULL)
   {
      printf("Error opening file for writing.\n");
      return 1;
   }

   fprintf(file, "Hello all. Welcome to Amrita\n");
   fclose(file);

   char buffer[100];
   file = fopen(filename, "r");
   if (file == NULL)
   {
      printf("Error opening file for reading.\n");
      return 1;
   }
   printf("Contents of the file:\n");
   fgets(buffer, sizeof(buffer), file);
```

```
    printf("%s", buffer);

    fclose(file);
    return 0;
}
```